

Bitwise linear trail through **LBlock** found by SAT

CiVerLy

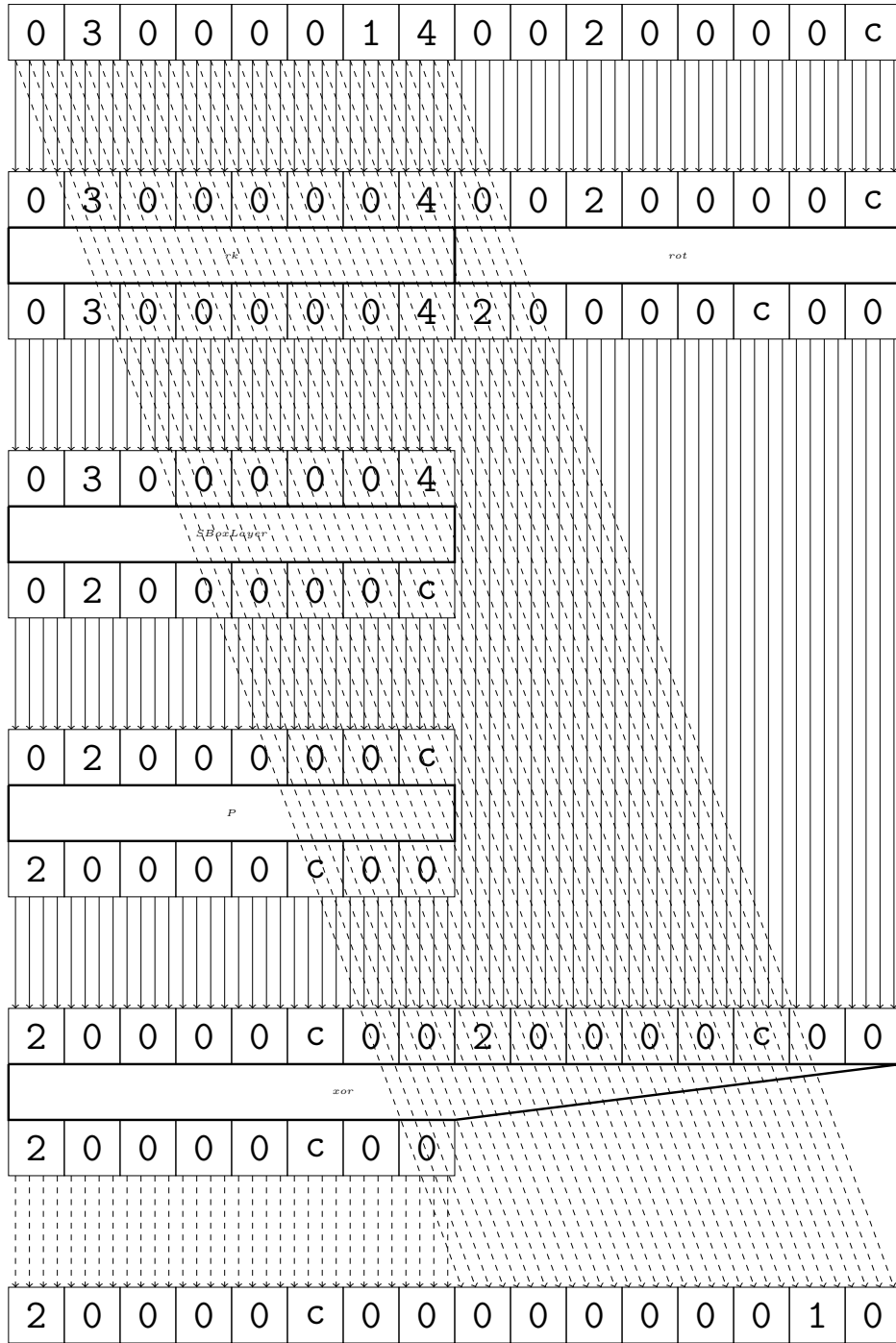
July 20, 2026

1 LBlock

Maximal linear correlation: 2^{-11}

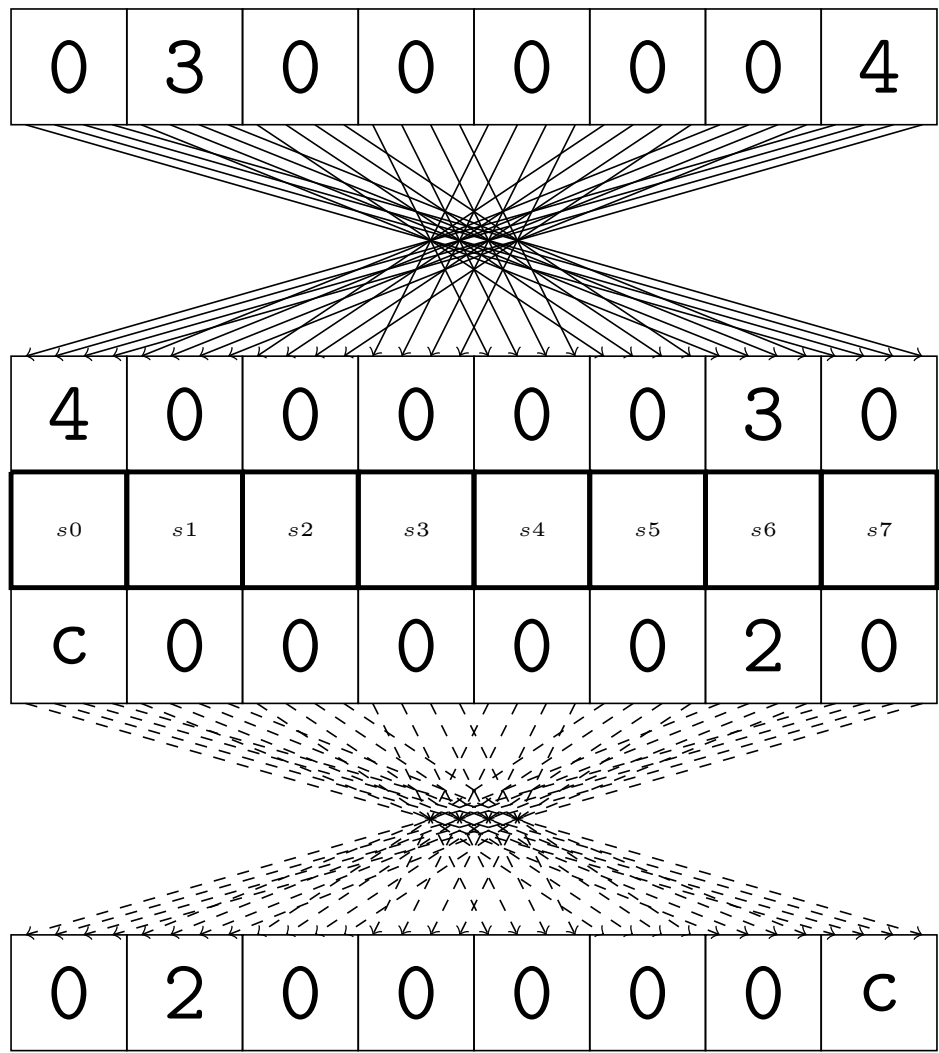
0	3	0	0	0	0	1	4	0	0	2	0	0	0	0	c
0	3	0	0	0	0	1	4	0	0	2	0	0	0	0	c
LBlockround															
2	0	0	0	0	0	c	0	0	0	0	0	0	0	1	0
2	0	0	0	0	0	c	0	0	0	0	0	0	0	1	0
LBlockround															
0	0	0	0	1	0	0	0	2	0	0	0	0	0	0	0
0	0	0	0	1	0	0	0	2	0	0	0	0	0	0	0
LBlockround															
0	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0
LBlockround															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	0
LBlockround															
0	0	0	0	2	0	0	0	0	0	0	0	0	7	0	0
0	0	0	0	2	0	0	0	0	0	0	0	0	7	0	0
LBlockround															
0	0	0	7	0	0	0	0	0	0	b	0	2	0	0	0
0	0	0	7	0	0	0	0	0	0	b	0	2	0	0	0
LBlockround															
b	0	2	0	0	0	0	0	b	5	0	7	0	0	0	0
b	0	2	0	0	0	0	0	b	5	0	7	0	0	0	0
LBlockround															
0	7	0	0	0	0	b	5	b	0	2	c	c	0	4	0
0	7	0	0	0	0	b	5	b	0	2	c	c	0	4	0
FinalSwap															
b	0	2	c	c	0	4	0	0	7	0	0	0	0	b	5
b	0	2	c	c	0	4	0	0	7	0	0	0	0	b	5

2 LBlock_round

Maximal linear correlation: $2^{-2.0}$ 

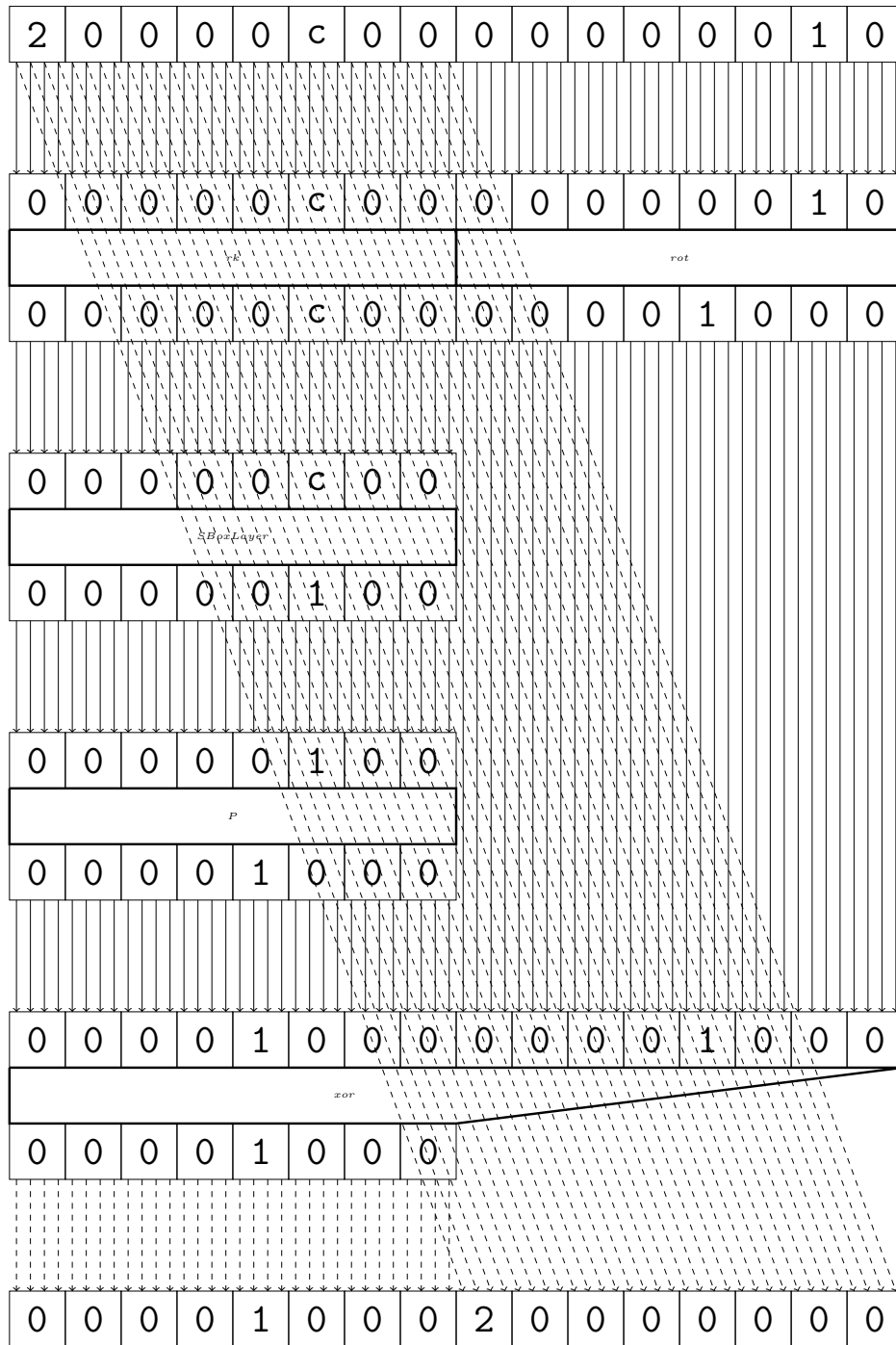
3 SBoxLayer

Maximal linear correlation: $2^{-2.0}$



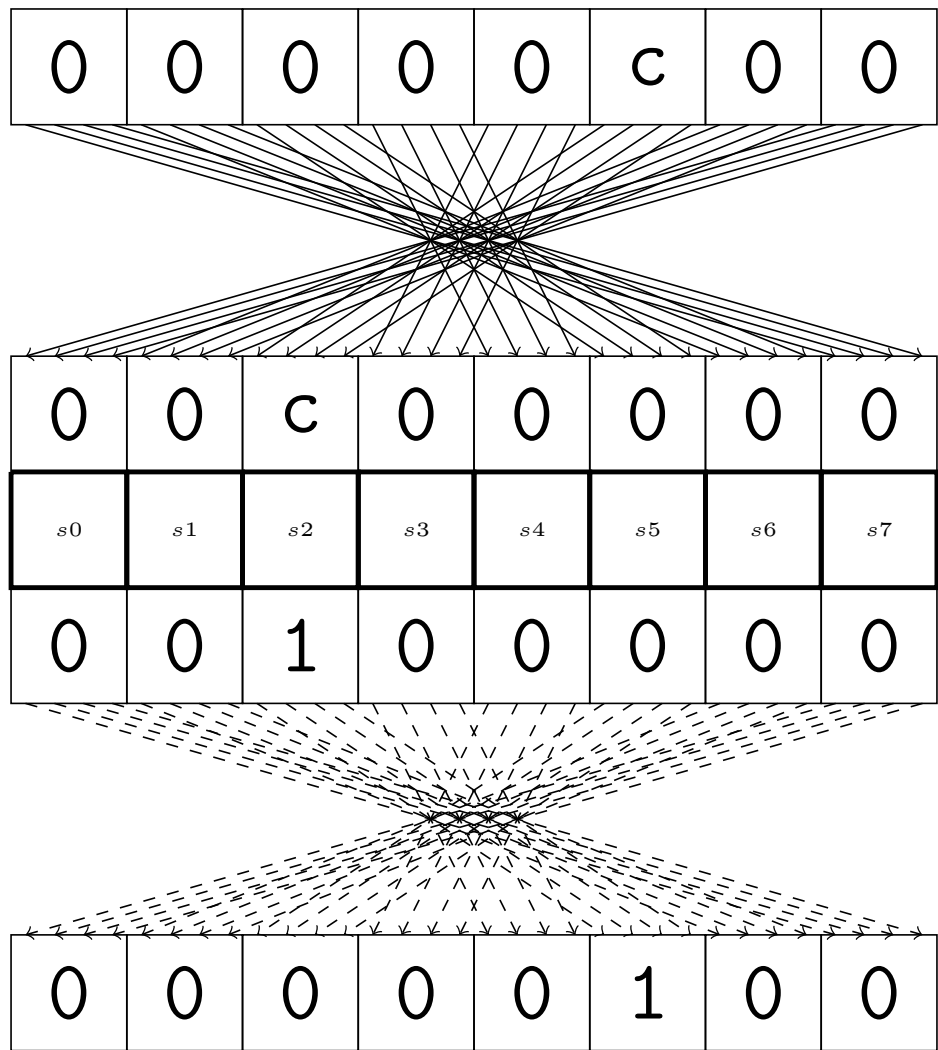
4 LBlock_round

Maximal linear correlation: $2^{-1.0}$



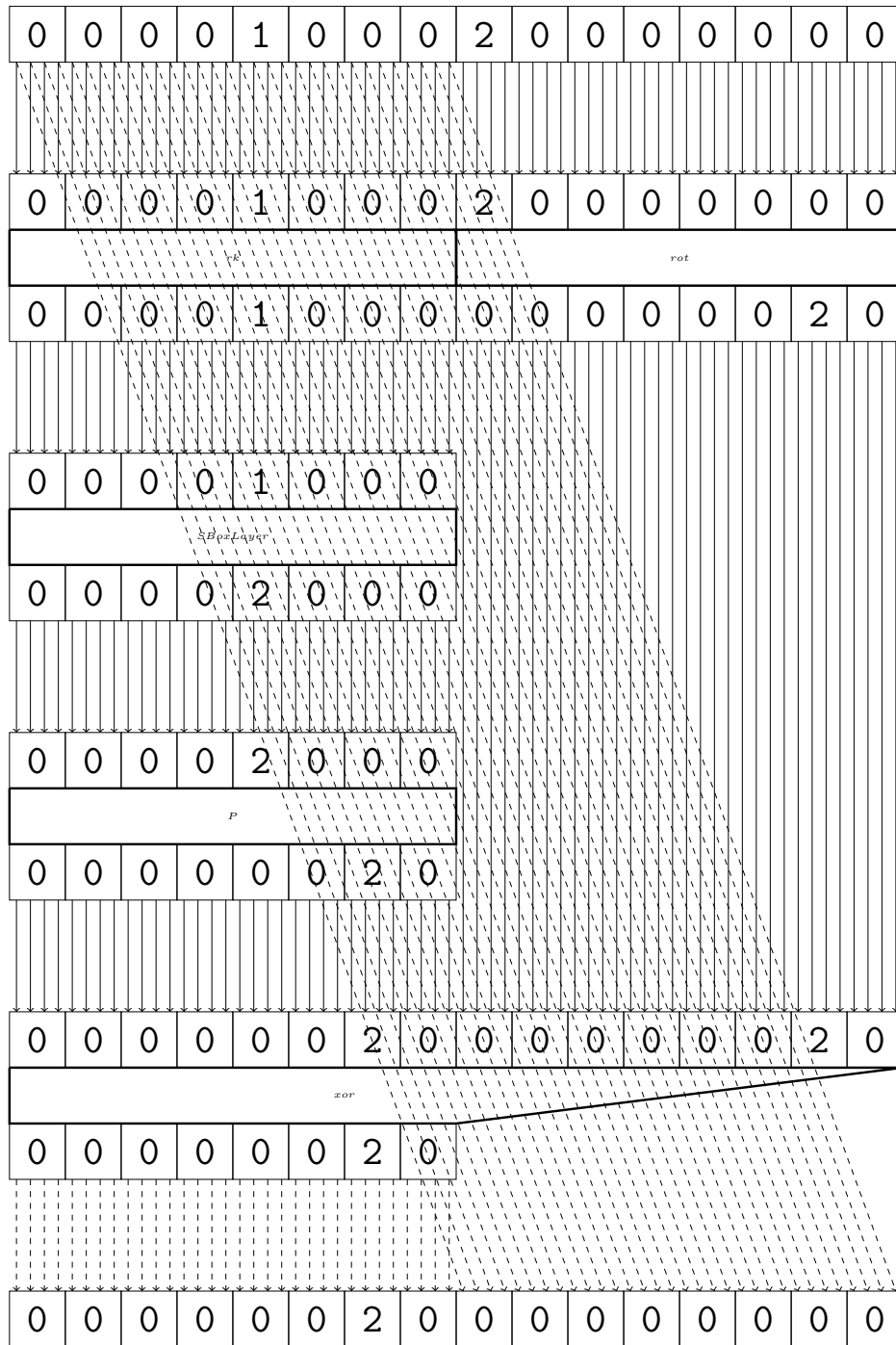
5 SBoxLayer

Maximal linear correlation: $2^{-1.0}$



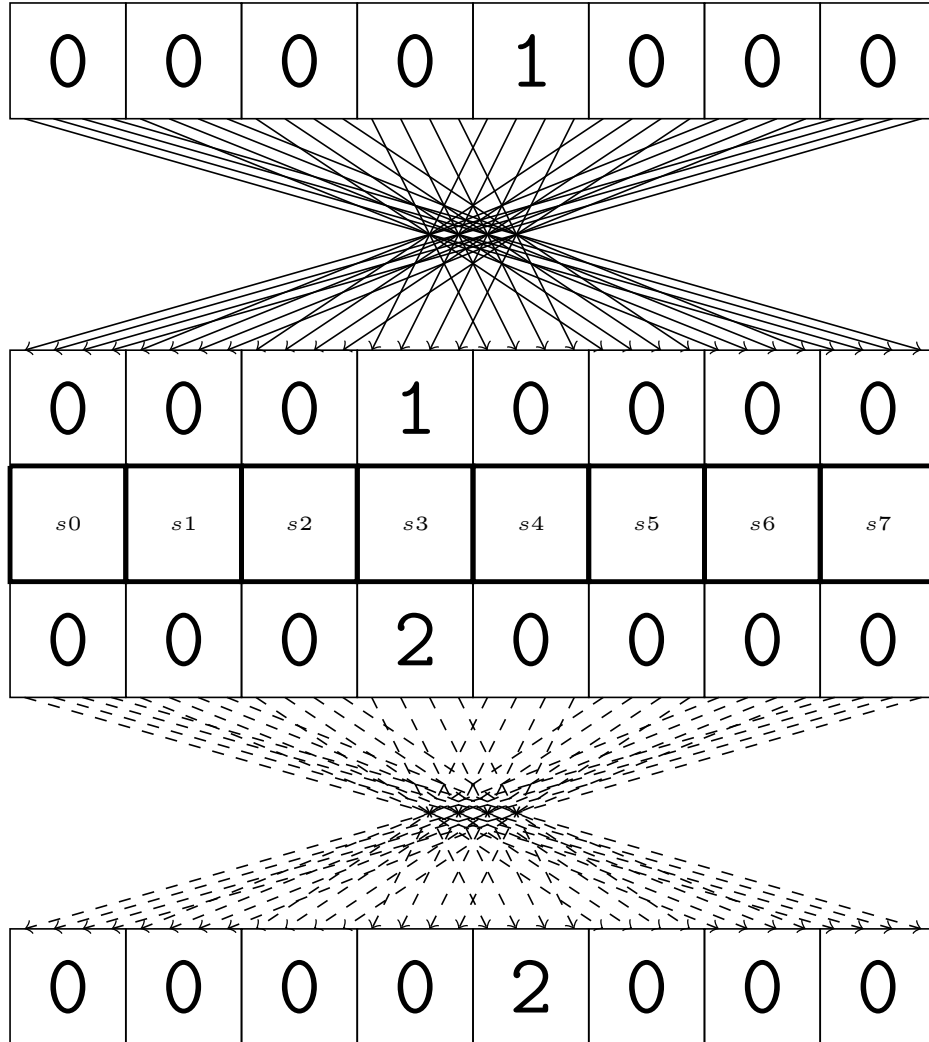
6 LBlock_round

Maximal linear correlation: $2^{-1.0}$



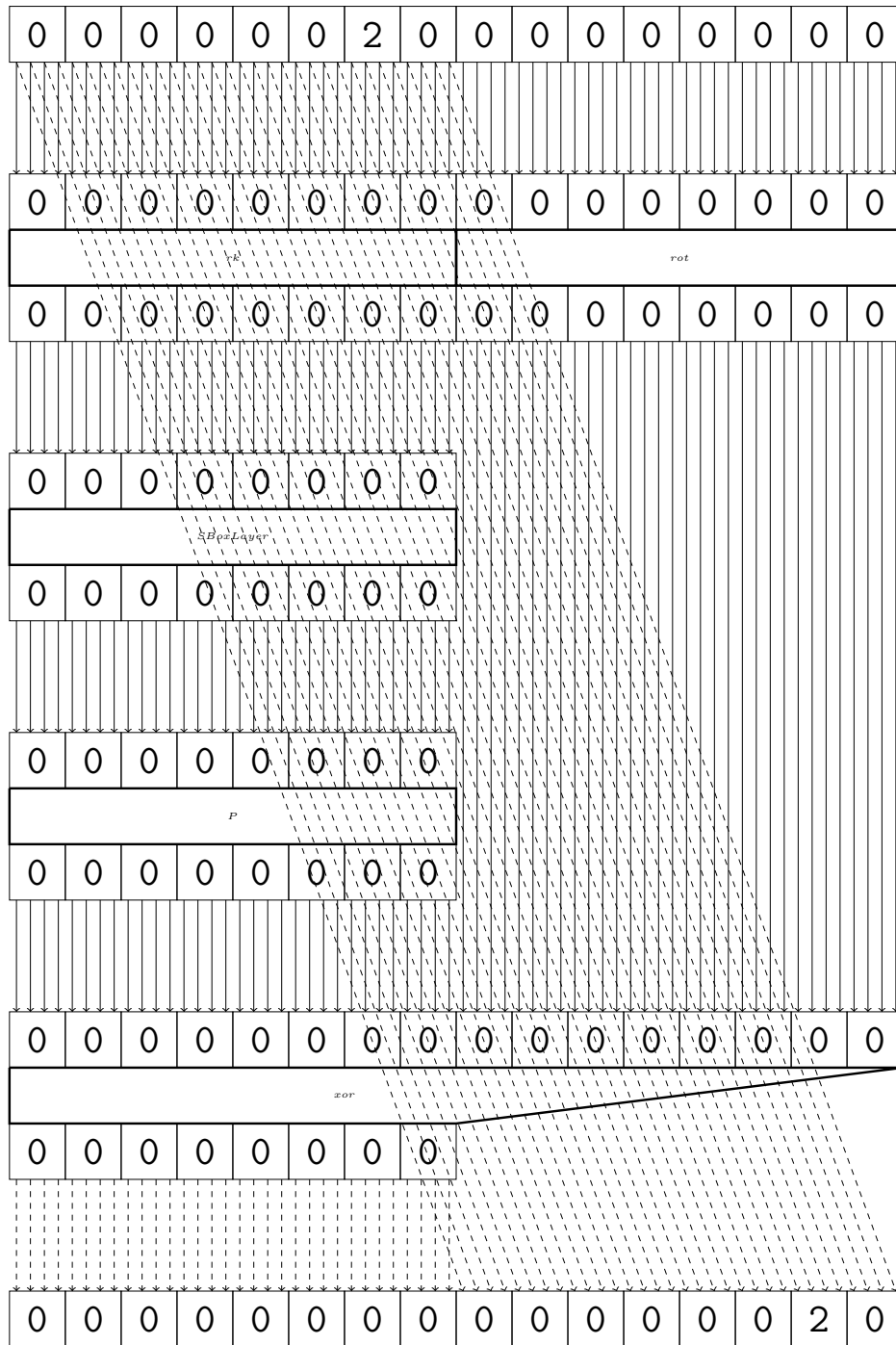
7 SBoxLayer

Maximal linear correlation: $2^{-1.0}$



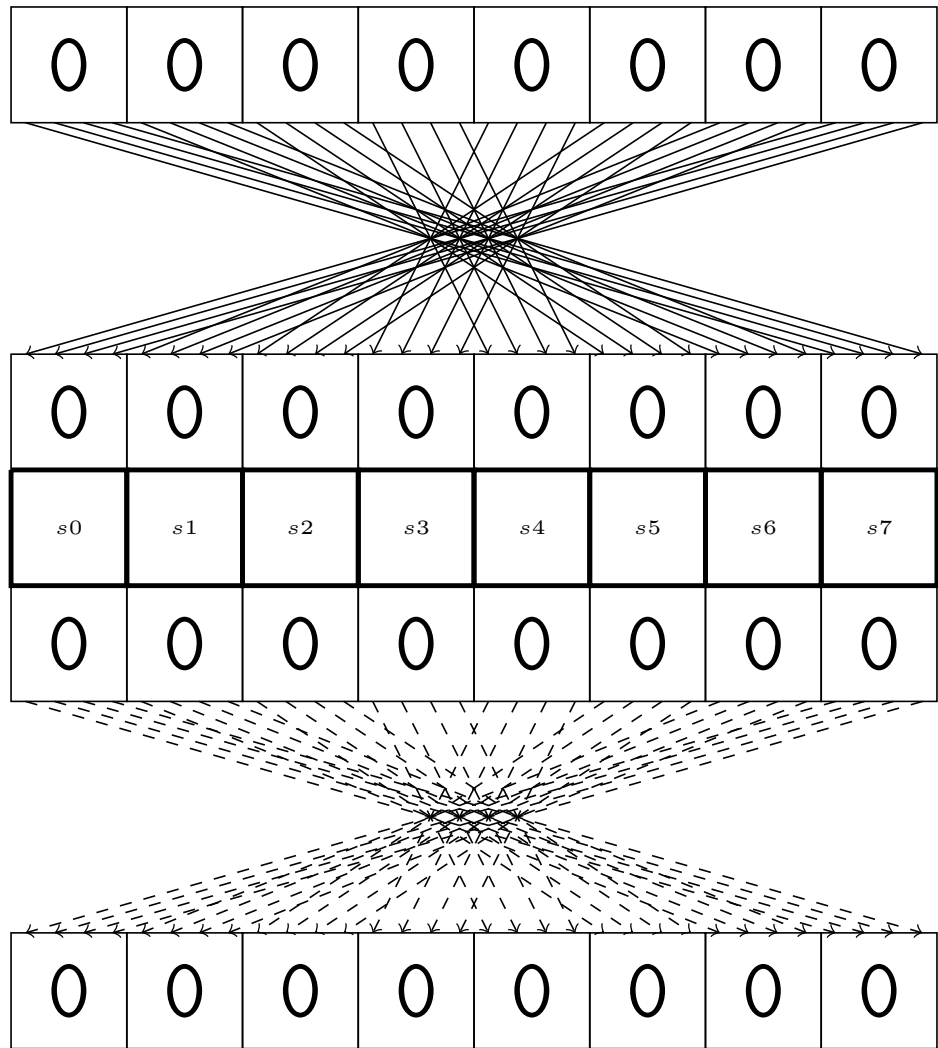
8 LBlock_round

Maximal linear correlation: $2^{-0.0}$

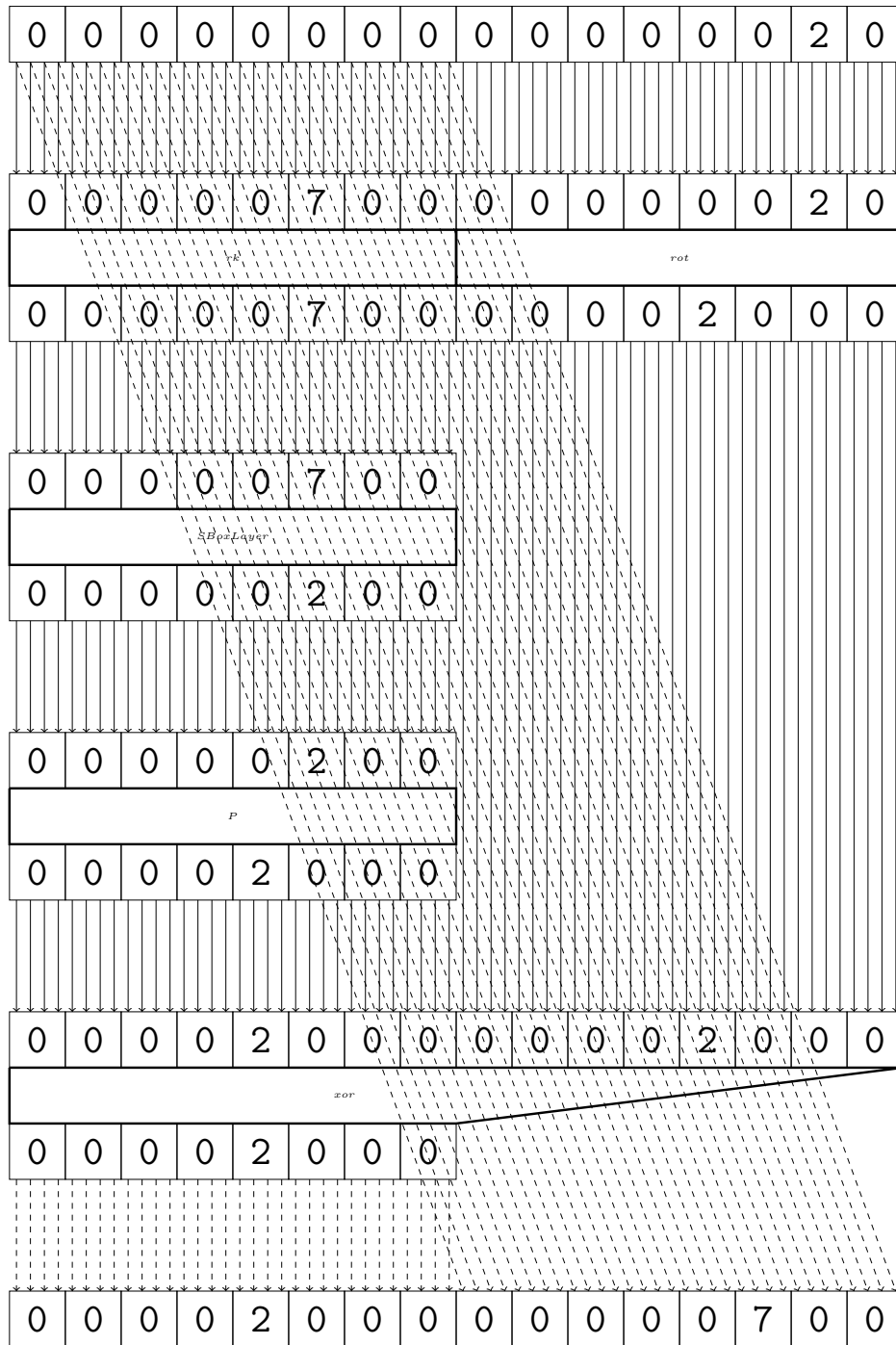


9 SBoxLayer

Maximal linear correlation: $2^{-0.0}$

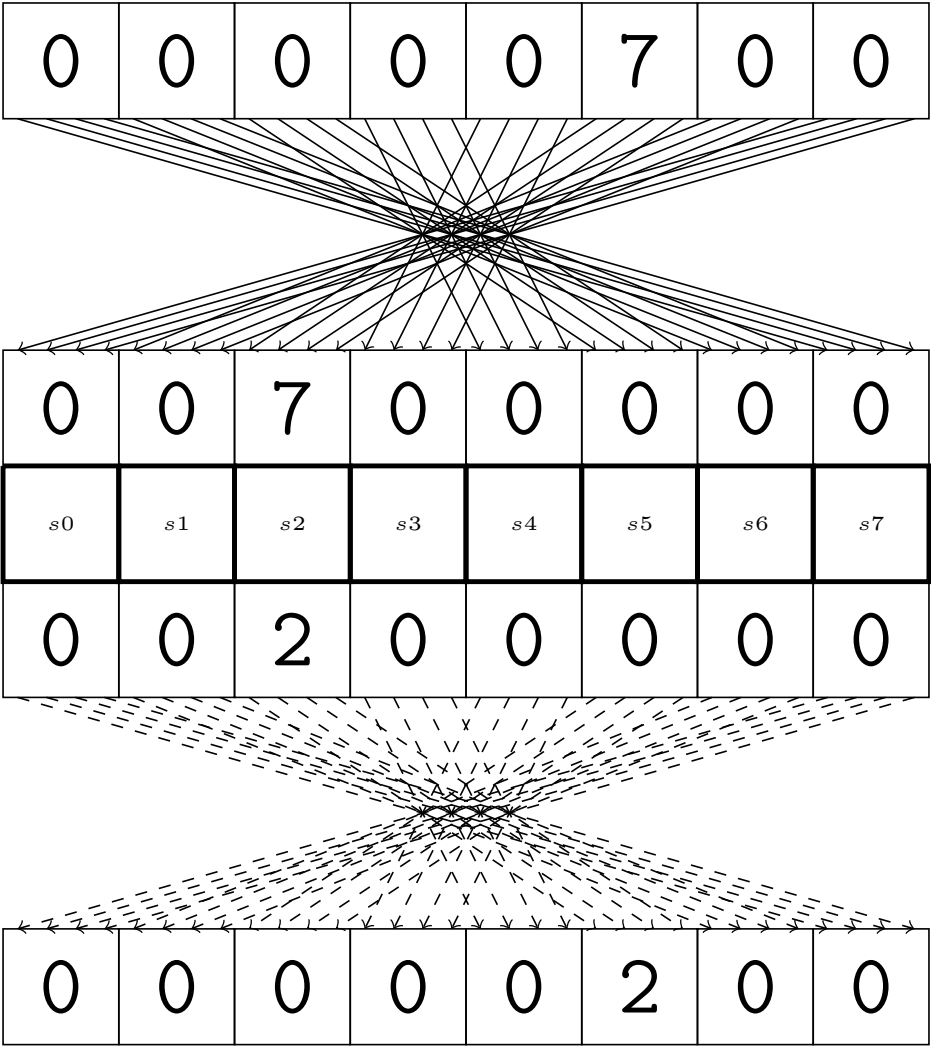


10 LBlock_round

Maximal linear correlation: $2^{-1.0}$ 

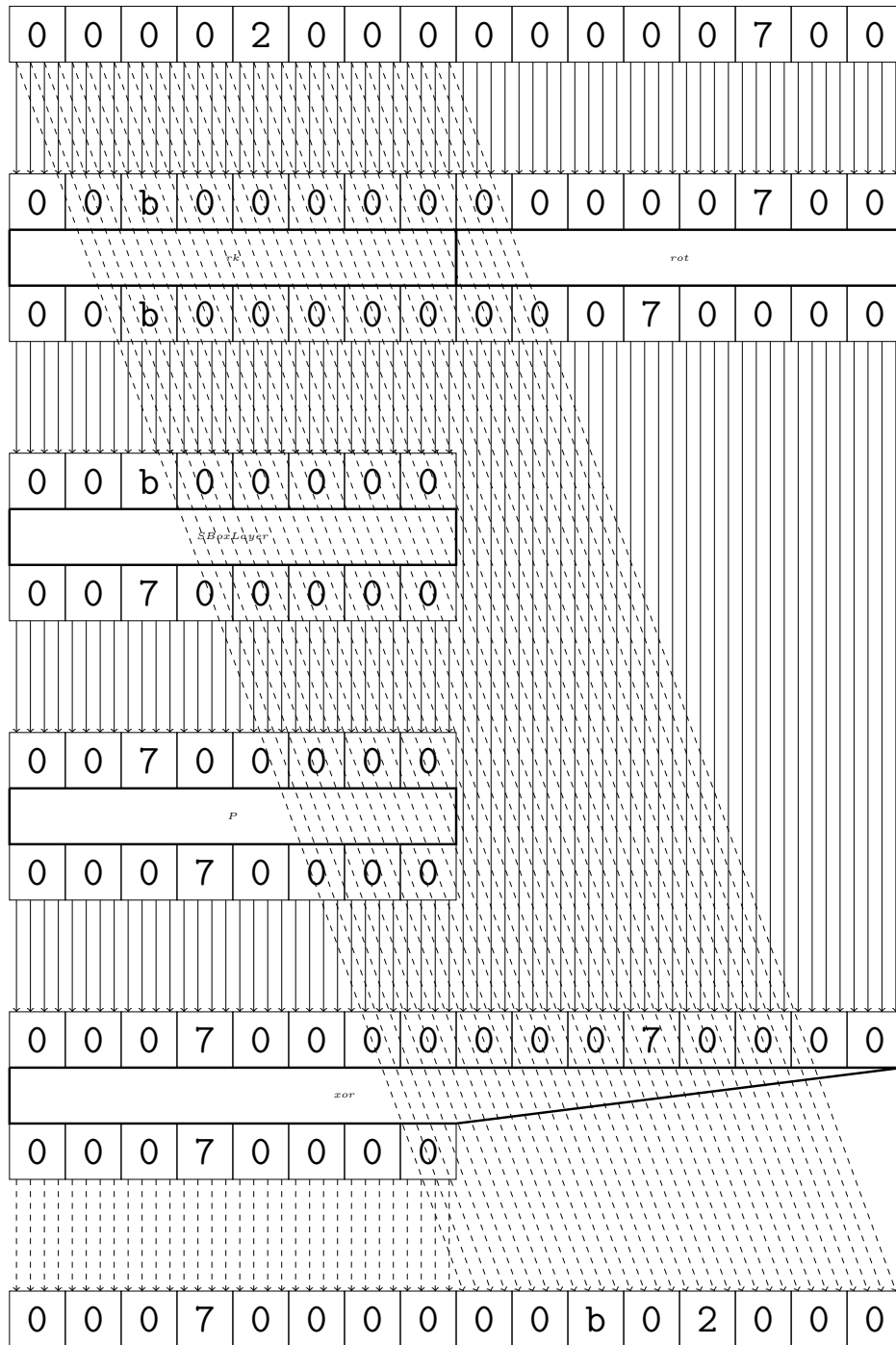
11 SBoxLayer

Maximal linear correlation: $2^{-1.0}$



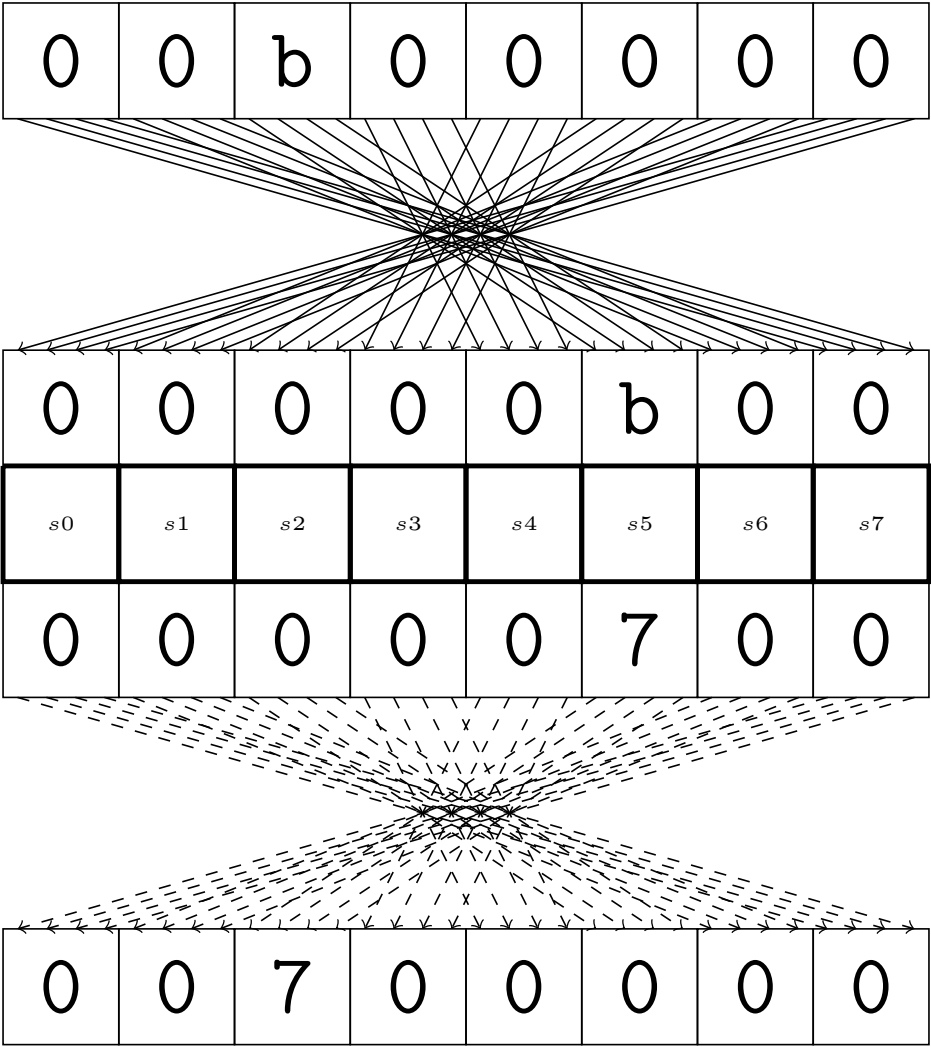
12 LBlock_round

Maximal linear correlation: $2^{-1.0}$



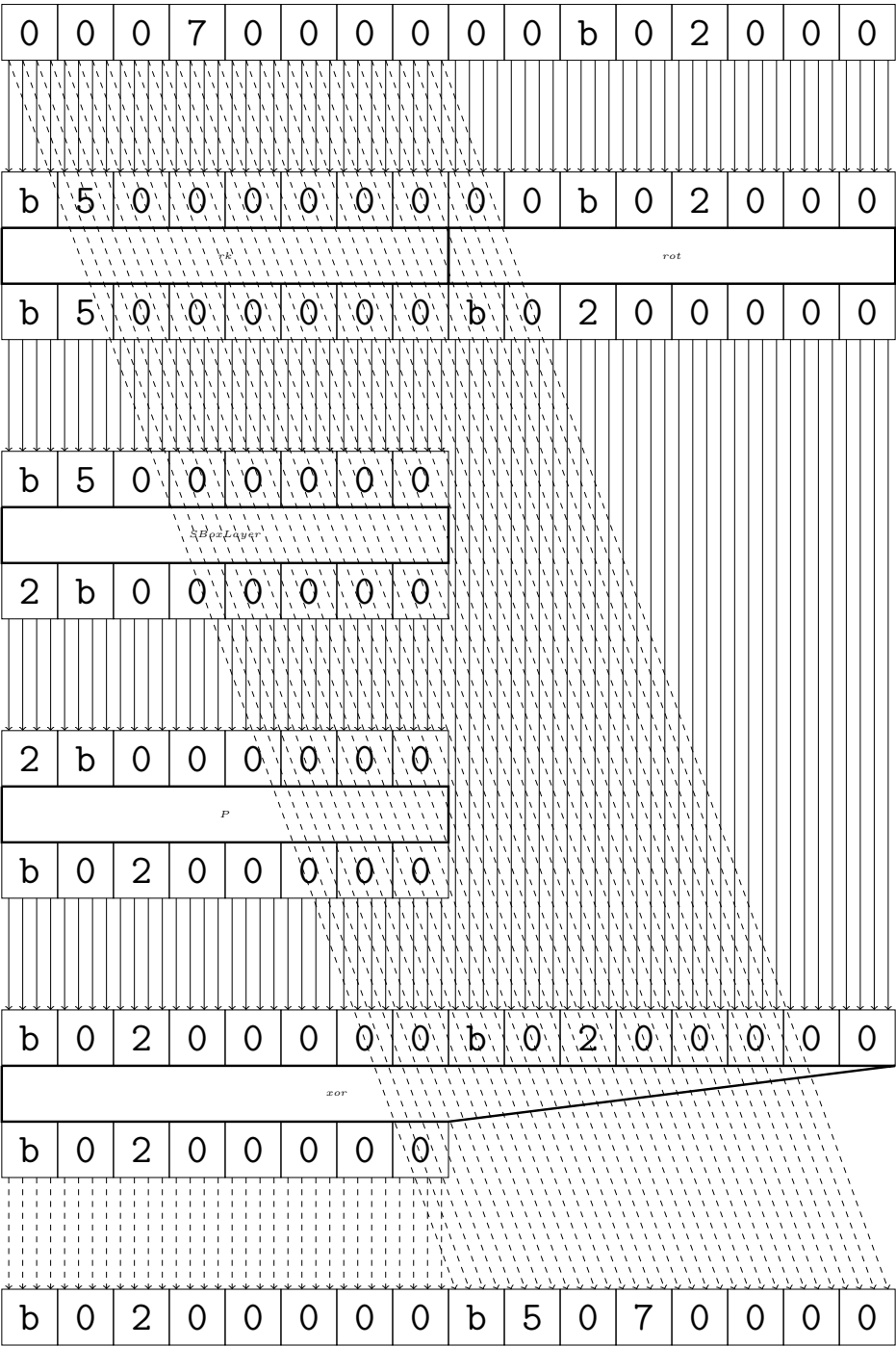
13 SBoxLayer

Maximal linear correlation: $2^{-1.0}$



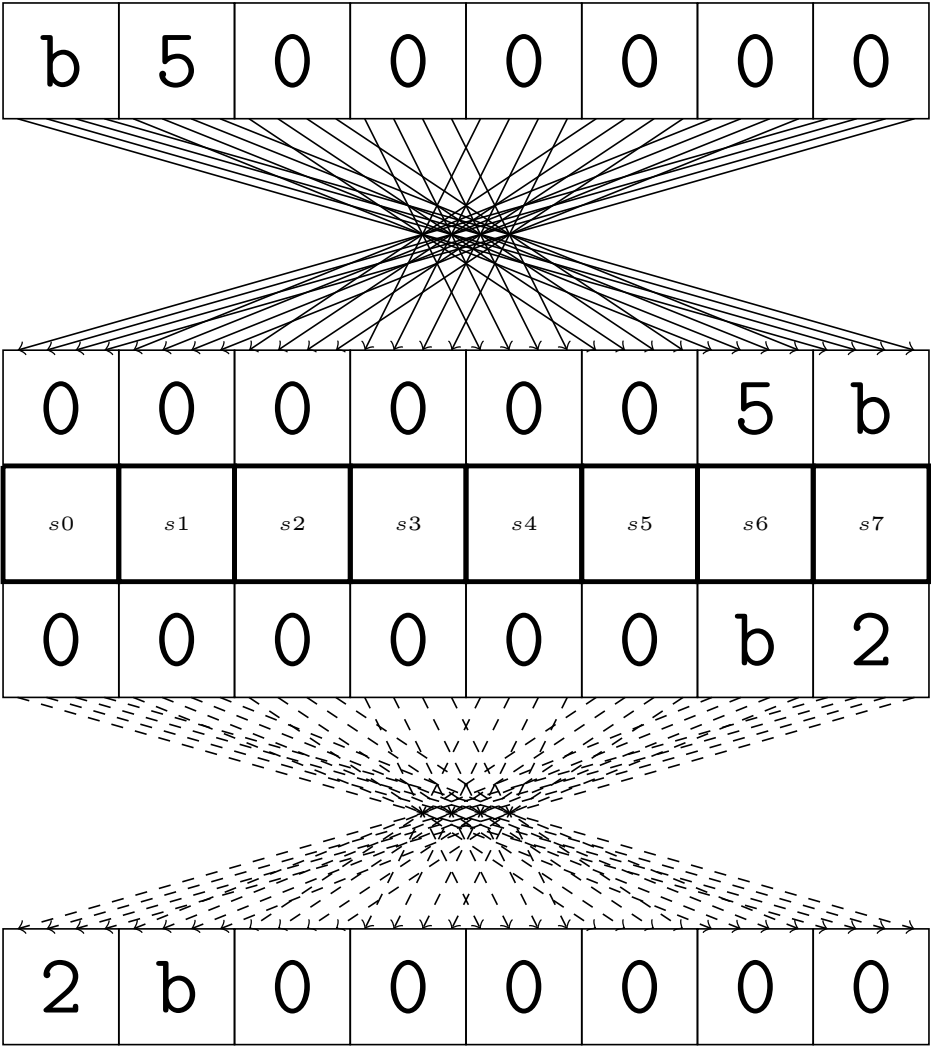
14 LBlock_round

Maximal linear correlation: $2^{-2.0}$

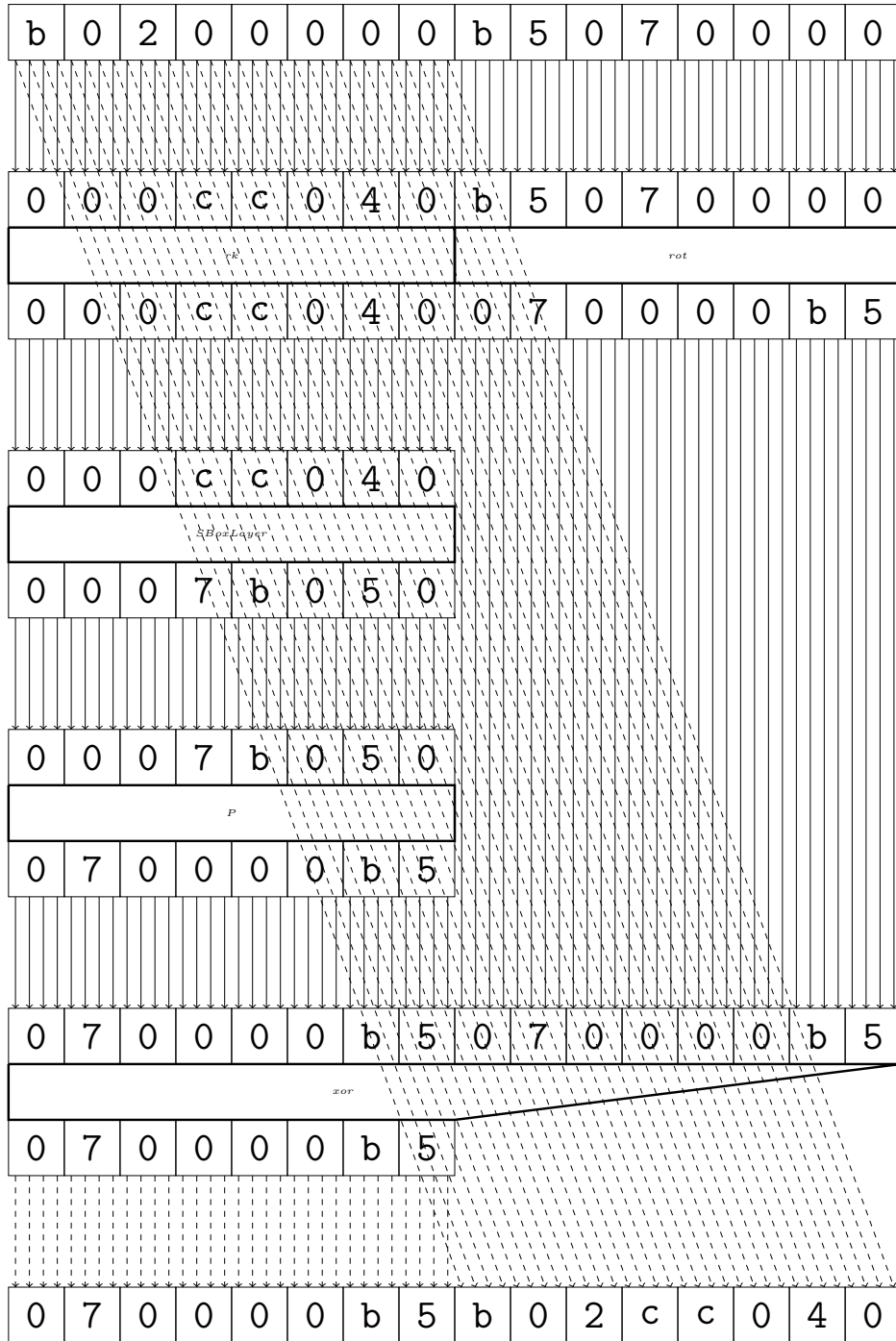


15 SBoxLayer

Maximal linear correlation: $2^{-2.0}$



16 LBlock_round

Maximal linear correlation: $2^{-3.0}$ 

17 SBoxLayer

Maximal linear correlation: $2^{-3.0}$

